

Windows Setup: Node.js, npm & Jest (PowerShell)

A student-friendly guide to set up testing for your JavaScript assignments—no WSL required.



What You'll Do

In about 10–15 minutes, you will:

- Install a modern version of Node.js using *nvm-windows* (a simple version manager).
- Set up a single shared Jest install for *all* your assignment folders.
- Run tests from PowerShell with a single command: **npm test**.
- Optionally use VS Code to run individual tests from the editor.

Why not WSL?

You don't need Linux/WSL for this course. PowerShell + Node is perfect for web dev and Jest-based testing. You can always add WSL later if you start doing Linux shell scripting.

A) Install Node the Clean Way (nvm-windows)

We'll use **nvm-windows** so you can easily install and switch Node versions without admin headaches.

1) Install nvm-windows

Open PowerShell and run:

```
winget install CoreyButler.NVMforWindows
```

Close and reopen PowerShell after the installer finishes.

2) Install and select Node 20 (recommended LTS)

```
nvm install 20
nvm use 20
node -v
npm -v
```

You should see Node v20.x and a modern npm (9 or 10).

B) One Shared Jest for All Assignments

Keep all your assignments under a single parent folder and install Jest *once* at the top level.

```
NeuroDev\  
  ■■ WebDev-II-Assignments\  
  ■■ package.json <-- Jest lives here (devDependency)  
  ■■ node_modules\  
  ■■ WebDev-II-2.3-Loops\  
  ■ ■■ practice.js  
  ■ ■■ __tests__\loops.test.js  
  ■■ WebDev-II-2.4-NextLesson\ ...
```

From PowerShell, at the parent folder:

```
cd path\to\NeuroDev\WebDev-II-Assignments  
npm init -y  
npm install --save-dev jest
```

Add a root test script in **WebDev-II-Assignments/package.json**:

```
{  
  "name": "webdev-ii-assignments",  
  "private": true,  
  "scripts": {  
    "test": "jest"  
  },  
  "devDependencies": {  
    "jest": "^30.2.0"  
  }  
}
```

Now you can run tests:

```
# Run all tests under the parent folder  
npm test  
  
# Run tests for one lesson folder  
npm test -- WebDev-II-2.3-Loops
```

Per-Folder Shortcut (Optional)

If you want to run **npm test** from inside a lesson folder without installing Jest there, point the script to the root Jest:

In **WebDev-II-2.3-Loops/package.json**:

```
{  
  "name": "WebDev-II-2.3-Loops",  
  "scripts": {  
    "test": "node ../../node_modules/jest/bin/jest.js"  
  }  
}
```

Then inside that lesson folder:

```
npm run test
```

Bonus: npm Workspaces (Optional)

Workspaces let the parent manage multiple subprojects. At the root **package.json**:

```
{
  "name": "webdev-ii-assignments",
  "private": true,
  "workspaces": ["WebDev-II-2.3-Loops", "WebDev-II-2.4-NextLesson"],
  "scripts": {
    "test": "jest"
  },
  "devDependencies": {
    "jest": "^30.2.0"
  }
}
```

Then you can:

```
# Run all workspace tests
npm test

# Run tests in one workspace
npm test -w WebDev-II-2.3-Loops
```

C) VS Code: Run Tests from the Editor

Install one of these extensions for a smoother workflow:

- **Jest** (Orta): runs tests and shows inline results
- **Jest Runner**: right-click a test or describe() → Run/Debug

Tip: Make sure VS Code is using the same Node version you set with nvm-windows.

D) Alternatives (Use if You Prefer)

1) Global Jest (simple, but less reproducible)

```
npm install -g jest
jest --version

# In any folder with tests
npm test # if your package.json has "test": "jest"
```

Pro: one-time install. Con: global versions can drift between machines.

2) On-Demand with npx (zero local install)

```
npx jest

# Pin a major version
npx jest@30
```

Pro: no install per project. Con: needs internet the first time.

3) WSL (optional)

You don't need WSL for this course. Consider it only if you later do Linux shell scripting or packages that assume Linux. For web dev + Jest, Windows + PowerShell is perfect.

E) Troubleshooting Cheatsheet

"jest: not found"

Install Jest at the parent (**npm i -D jest**) or globally (**npm i -g jest**).

If using a per-folder script, verify the path to **../node_modules/jest/bin/jest.js**.

Node is too old

Run **node -v**. If it's < **18**, install a modern Node:

nvm install 20 then **nvm use 20**.

PowerShell can't see nvm or node

Close and reopen PowerShell so your PATH updates, then run **nvm use 20** again.

Run tests for a single file or folder

Examples:

npm test -- WebDev-II-2.3-Loops

npx jest path\to\file.test.js

Quick Reference

```
# Install Node 20 (Windows)
winget install CoreyButler.NVMforWindows
nvm install 20
nvm use 20

# One shared Jest at the assignments root
cd path\to\NeuroDev\WebDev-II-Assignments
npm init -y
npm install --save-dev jest

# Run tests
npm test # all tests
npm test -- WebDev-II-2.3-Loops
npx jest # on demand (no local install)
```

You're set! If you get stuck, check the Troubleshooting section or ask your mentor.